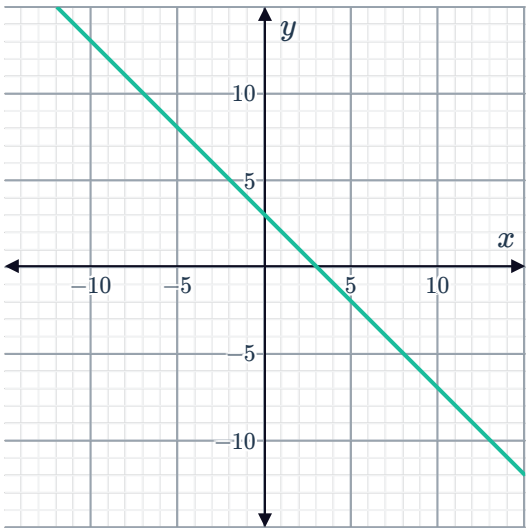
Constant rates of change

1 State whether the following functions have a constant or variable rate of change:

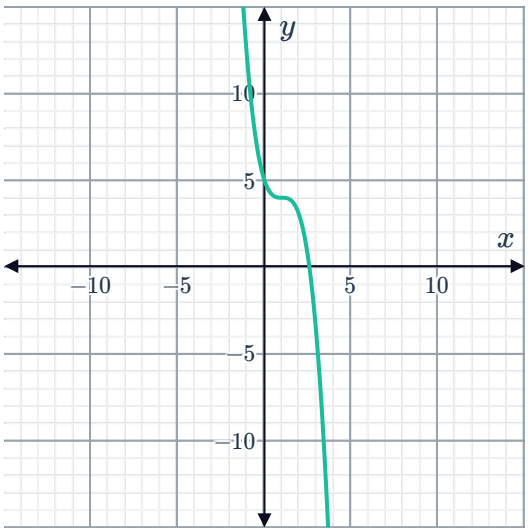
a $2x + y + 3 = 0$

b $y = 2^x$

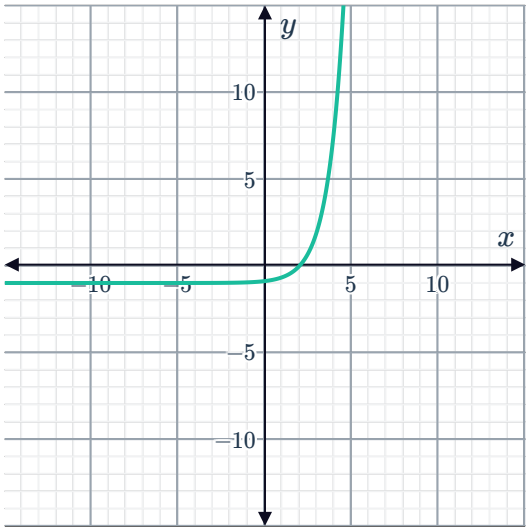
c



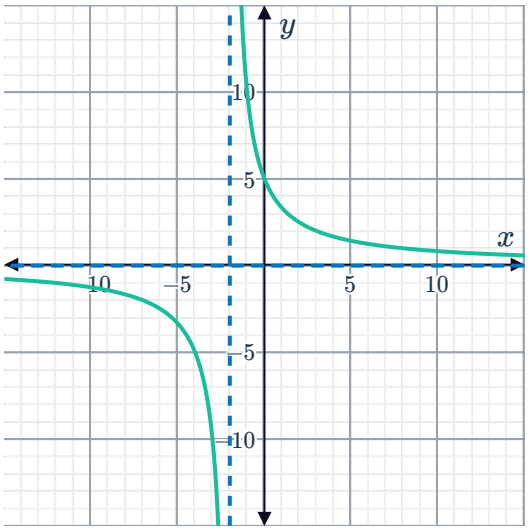
d



e



f



2 Does the function that passes through the following points:

$$\{(-2,-5),(1,-20),(2,-25),(7,-50),(9,-60)\}$$

have a constant or a variable rate of change?

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Average rates of change

3 Consider the function represented in the table:

x	-3	0	2	6
y	-4	-16	-22	-50

a Find the rate of change between $x = -3$ and $x = 0$.

b Find the rate of change between $x = 0$ and $x = 2$.

c Find the rate of change between $x = 2$ and $x = 6$.

d Does the function have a constant or a variable rate of change?

4 Consider the function represented in the table:

x	3	6	8	13
y	-12	-15	-17	-22

a Find the average rate of change between $x = 3$ and $x = 6$.

b Find the average rate of change between $x = 6$ and $x = 8$.

c Find the average rate of change between $x = 8$ and $x = 13$.

d Do the set of points satisfy a linear or non-linear function?

5 Consider the function represented in the table. Do the set of points satisfy a linear or non-linear function?

x	3	8	10	11	15
y	1	16	16	20	20

6 Consider the function that passes through the following points:

$$\{(5, -13), (9, -29), (11, -37), (16, -57)\}$$

a Find the average rate of change between $(5, -13)$ and $(9, -29)$.

b Find the average rate of change between $(9, -29)$ and $(11, -37)$.

c Find the average rate of change between $(11, -37)$ and $(16, -57)$.

d Do the points satisfy a linear or non-linear function?

7 Calculate the average rate of change of the function $f(x) = 5 \sin 3x$ over the domain $\frac{\pi}{} < x < \frac{\pi}{}$

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8 Consider the function $y = 2^x$:

a Find the average rate of change over the interval $[0, 1]$.

b Find the average rate of change over the interval $[1, 2]$.

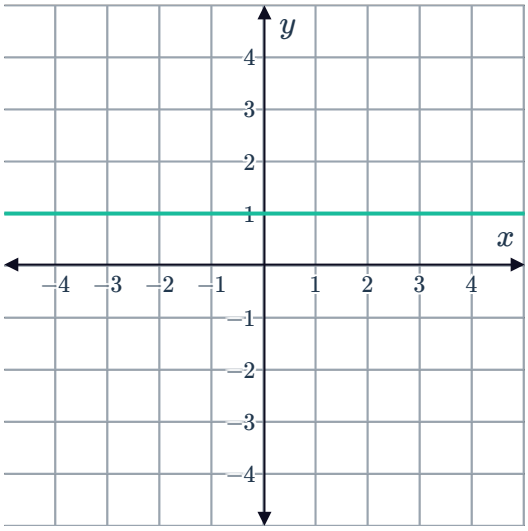
c Find the average rate of change over the interval $[2, 3]$.

d Hence, complete the following table:

Interval	$[3, 4]$	$[4, 5]$	$[5, 6]$
Average rate of change			

Instantaneous rates of change

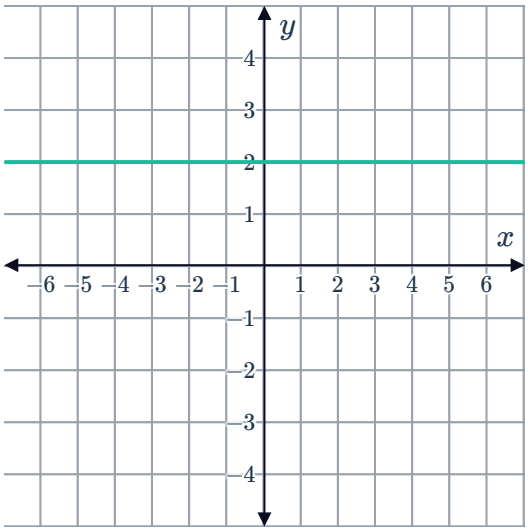
9 Consider the graph of the function $f(x) = 1$. If we were to draw a tangent to the function for any x-value, find the gradient of that tangent.



10 Consider the graph of the function $f(x) = 2$:

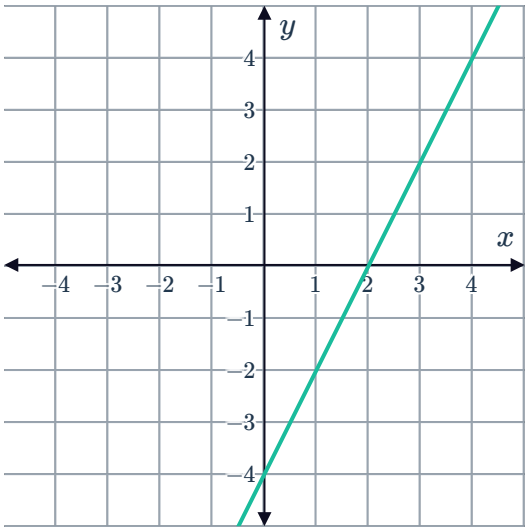
a Calculate the average rate of change between $x = 3$ and $x = 6$.

b Find the gradient of the tangent at the point $x = 3$.



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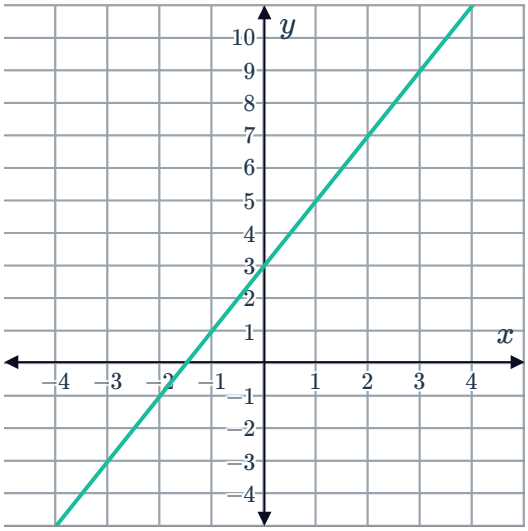
11 Consider the graph of the function $f(x) = 2x - 4$. If we were to draw a tangent to the function for any x-value, find the gradient of that tangent.



12 Consider the graph of the function $f(x) = 2x + 3$:

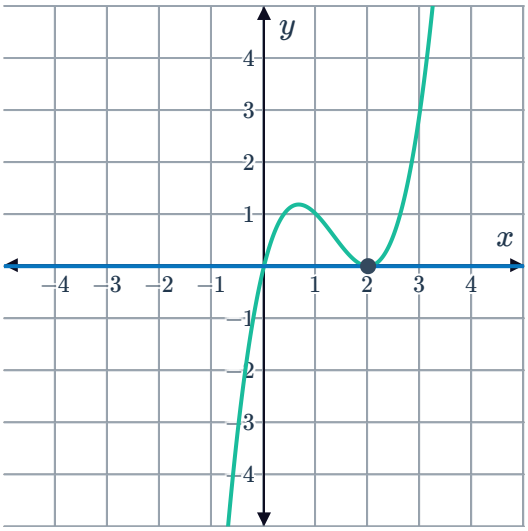
a Calculate the average rate of change between $x = -2$ and $x = 2$.

b Find the gradient of the tangent at the point $x = -2$.

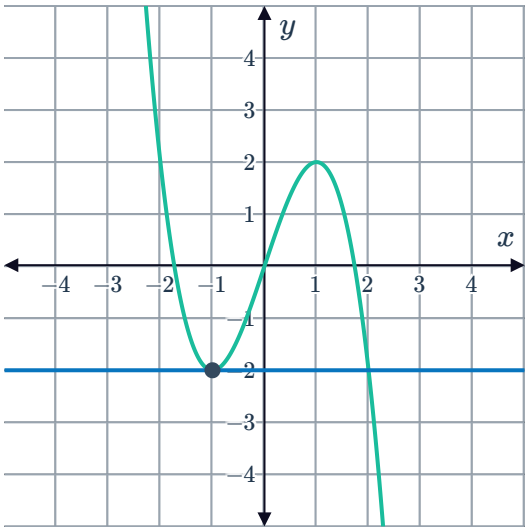


13 For each of the following graphs, find the gradient of the tangent at the given point, and hence the instantaneous rate of change:

a

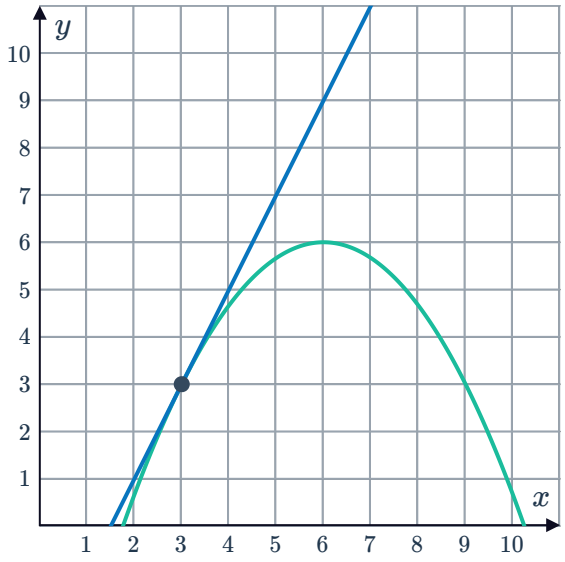


b

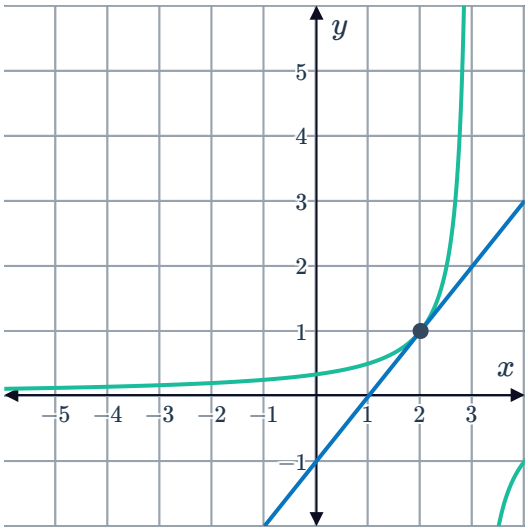


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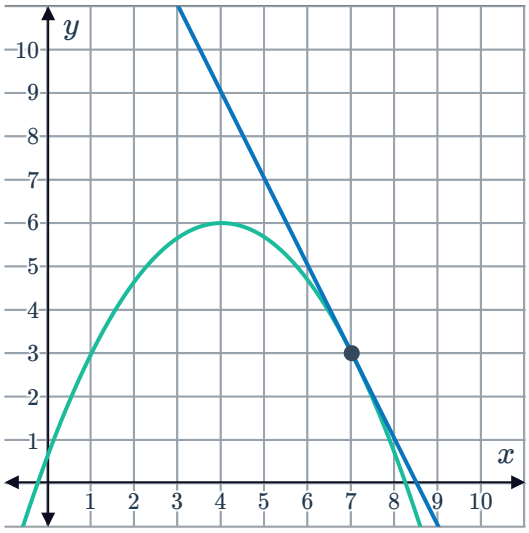
c



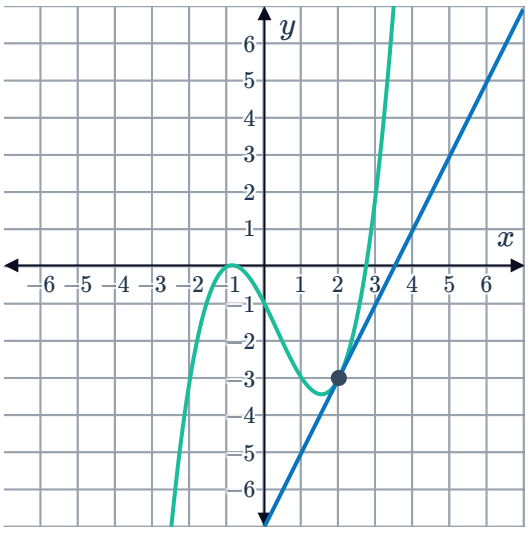
d



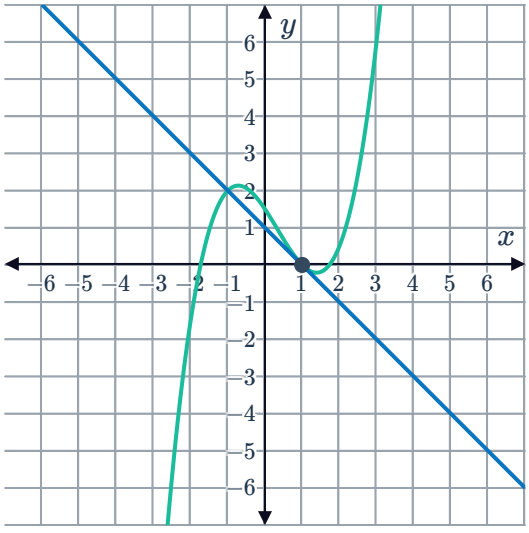
e



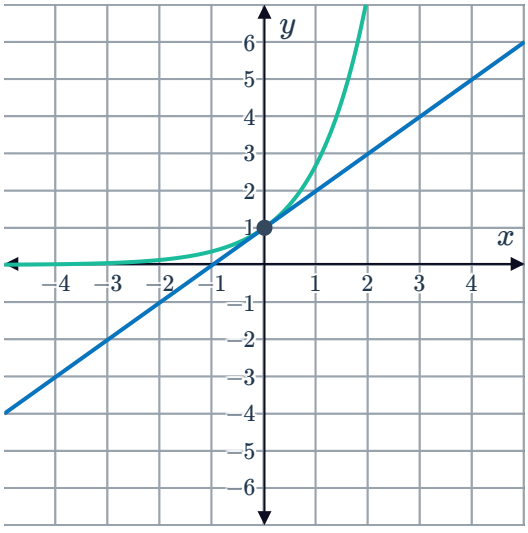
f



g

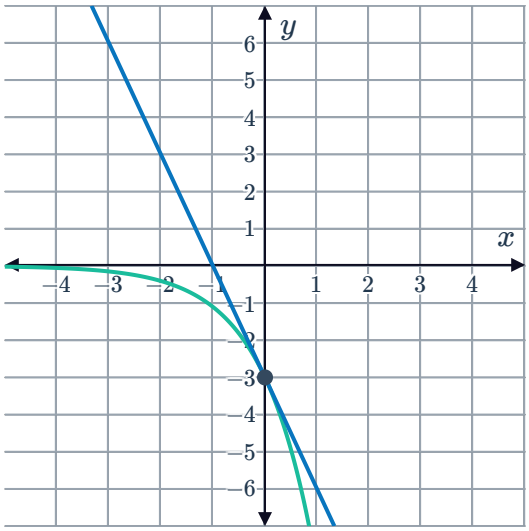


h



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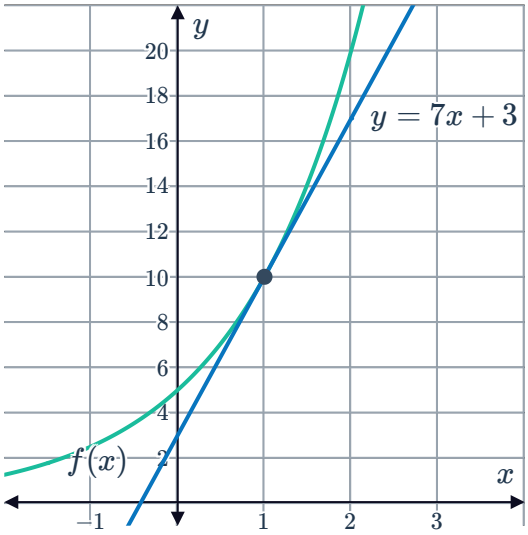
i



14 Consider the function $f(x) = 5(2)^x$ and the tangent line given by the equation $y = 7x + 3$ at $x = 1$:

a Calculate the average rate of change between $x = 1$ and $x = 2$.

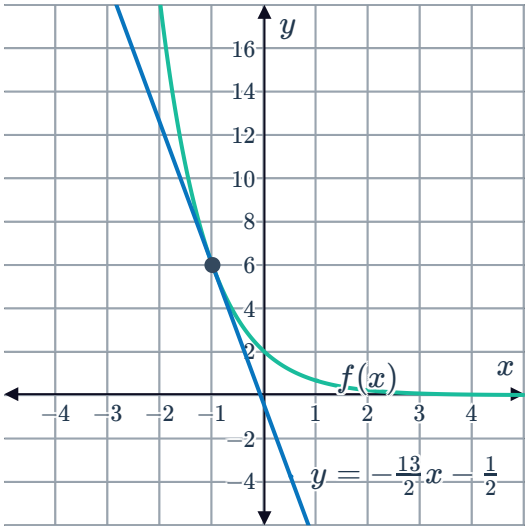
b Calculate the instantaneous rate of change at $x = 1$.



15 Consider the function $f(x) = 2(3)^{-x}$ and the tangent line given by the equation $y = -\frac{13}{2}x - \frac{1}{2}$ at $x = -1$:

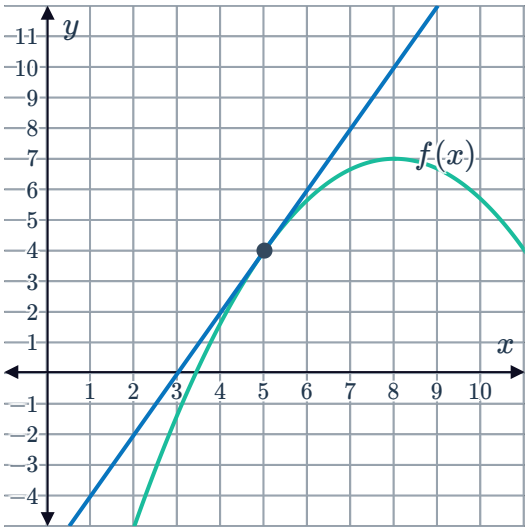
a Calculate the average rate of change between $x = -1$ and $x = 0$

b Calculate the instantaneous rate of change at $x = -1$.



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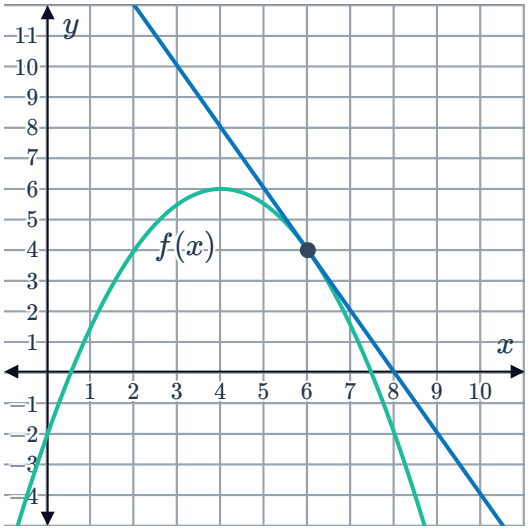
16 Consider the function $f(x) = -\frac{(x-8)^2}{3} + 7$ and a tangent line at $x = 5$:



a Calculate the average rate of change between $x = 5$ and $x = 8$

b Calculate the instantaneous rate of change at $x = 5$.

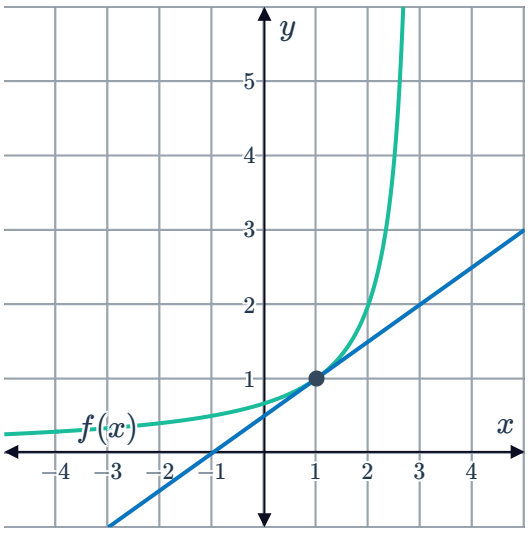
17 Consider the function $f(x) = -\frac{(x-4)^2}{2} + 6$ and a tangent line at $x = 6$:



a Calculate the average rate of change between $x = 6$ and $x = 8$.

b Calculate the instantaneous rate of change at $x = 6$.

18 Consider the function $f(x) = -\frac{2}{x-3}$ and a tangent line at $x = 1$:

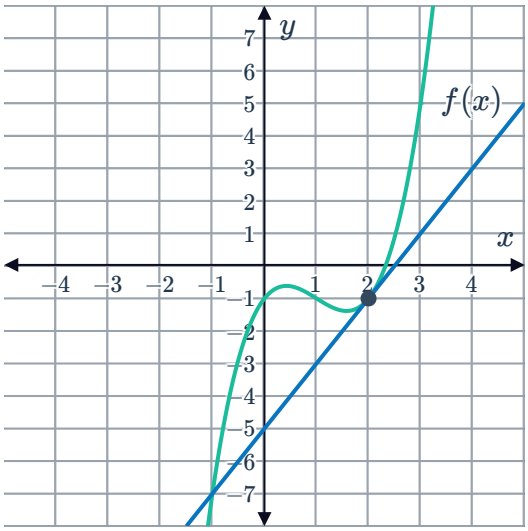


a Calculate the average rate of change between $x = 1$ and $x = \frac{3}{2}$

b Calculate the instantaneous rate of change at $x = 1$.

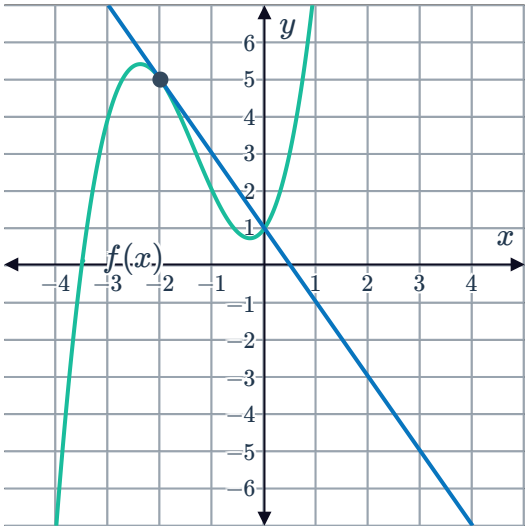
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19 Consider the function $f(x) = x^3 - 3x^2 + 2x - 1$ and a tangent line at $x = 2$:



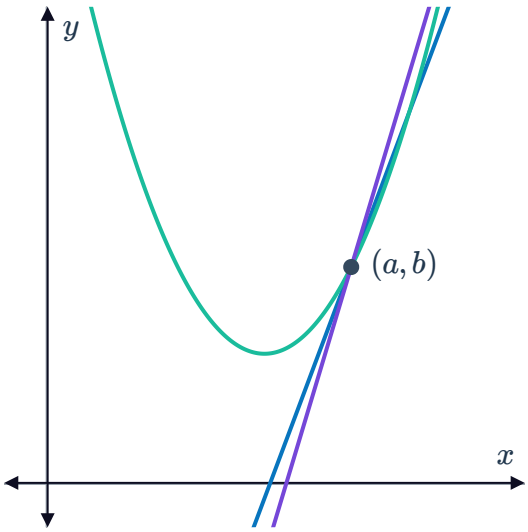
- **a** Calculate the average rate of change between $x = 2$ and $x = 3$.
- **b** Calculate the instantaneous rate of change at $x = 2$.

20 Consider the function $f(x) = x^3 + 4x^2 + 2x + 1$ and a tangent line at $x = -2$:



- **a** Calculate the average rate of change between $x = -2$ and $x = -1$.
- **b** Calculate the instantaneous rate of change at $x = -2$.

21 Consider the following figure. One secant line through the given point has the equation $y = 2.02x - 9.14$, while another secant line through the same point has the equation $y = 2.01x - 9.07$:



- **a** Find the value of a.
- **b** Find the value of b.
- **c** Find the gradient of the tangent at this point to the nearest integer.

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22 Consider the function $f(x) = x^3$.

a By filling in the table of values, complete the limiting chord process for $f(x) = x^3$ at the point $x = 1$. Round your answers to four decimal places if needed.

b Find the instantaneous rate of change of $f(x)$ at $x = 1$.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	
1	1.5		
1	1.1		
1	1.05		
1	1.01		
1	1.001		
1	1.0001		

23 Consider the function $f(x) = 2x^2$.

a By filling in the table of values, complete the limiting chord process for $f(x) = 2x^2$ at the point $x = 1$.

b Find the instantaneous rate of change of $f(x)$ at $x = 1$.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	
1	1.5		
1	1.1		
1	1.05		
1	1.01		
1	1.001		
1	1.0001		

24 Consider the function $f(x) = 4^x$.

a By filling in the table of values, complete the limiting chord process for $f(x) = 4^x$ at the point $x = 0$. Round your answers to four decimal places if needed.

b Find the instantaneous rate of change of $f(x)$ at $x = 0$ correct to three decimal places.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
0	1	1	
0	0.5		
0	0.1		
0	0.05		
0	0.01		
0	0.001		
0	0.0001		

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25 Consider the function $f(x) = -x^2 + 5$.

a By filling in the table of values, complete the limiting chord process for $f(x) = -x^2 + 5$ at the point $x = 1$.

b Find the instantaneous rate of change of $f(x)$ at $x = 1$.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	
1	1.5		
1	1.1		
1	1.05		
1	1.01		
1	1.001		
1	1.0001		

26 Consider the function $f(x) = \frac{5}{x}$.

a By filling in the table of values, complete the limiting chord process for $f(x) = \frac{5}{x}$ at the point $x = 1$. Round your answers to four decimal places if needed.

b Find the instantaneous rate of change of $f(x)$ at $x = 1$.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	
1	1.5		
1	1.1		
1	1.05		
1	1.01		
1	1.001		
1	1.0001		

Applications

27 The population of koalas in a particular area can be modelled by the equation $y = 10 + 2^x$, where x is the number of years from now.

a Find the value of y when $x = 0$.

b Find the value of y when $x = 4$.

c Hence, find the population's average rate of change over the interval $[0, 4]$.

28 The daily net profit of an upmarket restaurant can be modelled by the equation $y = -17x^2 + 493x$, where x is the number of customers.

a Find the value of y at $x = 0$.

b Find the value of y at $x = 13$.

c Hence, find the average rate of change in net profit over the interval $[0, 13]$.

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29 The value of a particular coin can be modelled by the equation $y = 80\,(2^x)$, where x is the number of years from now. Find the average rate of change of its value over the interval $[0, 4]$.

30 The path of water projected from a fountain can be modelled by the equation $y = -20x^2 + 120x$, where x is the horizontal distance from the nozzle and y is the height. Find the average rate of change of the water's height over the interval $[0, 3]$.

31 The government is concerned about the impact of a wind farm on the population of local birds and is seeking to impose a limit on the number of new turbines that may be built on the farm each year. It has proprosed two options to the operators of the wind farm:

- **Option A**: The initial number of turbines is 20, and this number can be increased by 4 per year.
- **Option B**: The initial number of turbines is also 20, but the number of turbines can be increased by 10% per year.

Let y be the number of turbines, x years from now.

a Form an equation that will model the number of turbines on the farm in x years under option A.

b Form an equation that will model the number of turbines on the farm in x years under option B.

c What will be the average rate of change in the number of turbines over the next 28 years under option A?

d What will be the average rate of change in the number of turbines over the next 28 years under option B? Round your answer correct to two decimal places.

e Under which option will average annual growth in the number of turbines over the next 28 years be lower?

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32 The table shows the price of a piece of sporting memorabilia over a number of years:

Year	1981	1986	1996	2010	2014
Price	$156	$176	$196	$280	$284

a Find the average annual rate of change in the price between:

i 1981 and 1986 **ii** 1986 and 1996

iii 1996 and 2010 **iv** 2010 and 2014

b During which interval was the average annual price change the greatest?

33 The table shows the price of an antique whiskey bottle over a number of years:

Year	1984	1985	1999	2010	2012
Price	$181	$176	$92	$59	$57

a Find the average annual rate of change in the price between:

i 1984 and 1985 **ii** 1985 and 1999

iii 1999 and 2010 **iv** 2010 and 2012

b During which interval was the average annual price change the greatest?

34 The table shows the share price of a company at several times since its start.

Year	0	8	13	26	35
Share Price	$61.50	$95.90	$96.90	$176.20	$163.60

a Find the average rate of change in the share price between years:

i 0 and 8 **ii** 8 and 13 **iii** 13 and 26 **iv** 26 and 35

b During which interval was the average annual share price change the greatest?

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35 The table shows Skye's progress through a four-hour ultramarathon.

Time (hours)	0	1	2	3	4
Distance from starting line (kilometers)	0	19	37	48	63

a Find Skye's average speed during:

i The first hour.

ii The second hour.

iii The third hour.

iv The fourth hour.

b During which hour was Skye travelling the fastest?

36 The table shows the linear relationship between the temperature ($^\circ F$) on a particular day and the net profit, in dollars of a store. Find the rate of change of the net profit.

Temperature	-4	-3	-1	3
Net profit	49	39	19	-21

37 Due to adverse market conditions, Tobias and Gwen have had to reduce the number of staff at their respective companies, which currently have 310 staff.
Tobias plans on reducing staff numbers by 41 each year for the next five years, while Gwen plans on reducing staff numbers by 8 next year, 16 in the year after next, 24 in the year after that, and so on.

Year	0	1	2	3	4	5
Tobia's staff						
Gwen's staff						

a Complete the table. Note that year 0 represents the current year.

b What will be the rate of change in the size of Tobias's staff between years 3 and 4?

c What will be the rate of change in the size of Gwen's staff between years 3 and 4?

d Is the function representing the size of Tobias's staff linear or non-linear? Explain your answer.

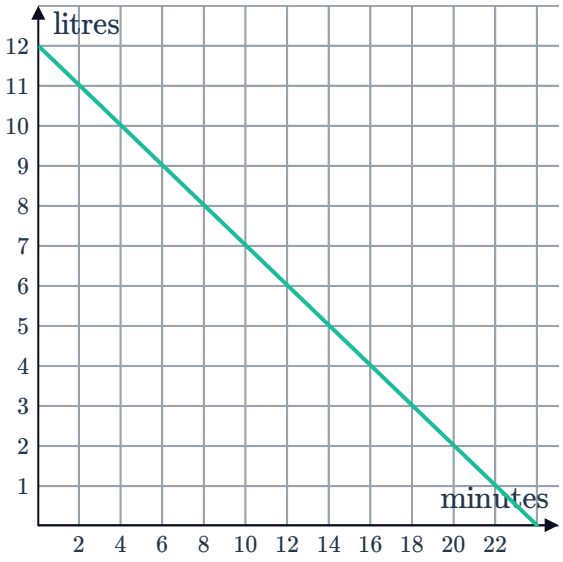
e Is the function representing the size of Gwen's staff linear or non-linear? Explain your answer.

f Whose staff will decrease more quickly between years 3 and 4?

g Whose staff will decrease more quickly between years 10 and 11?

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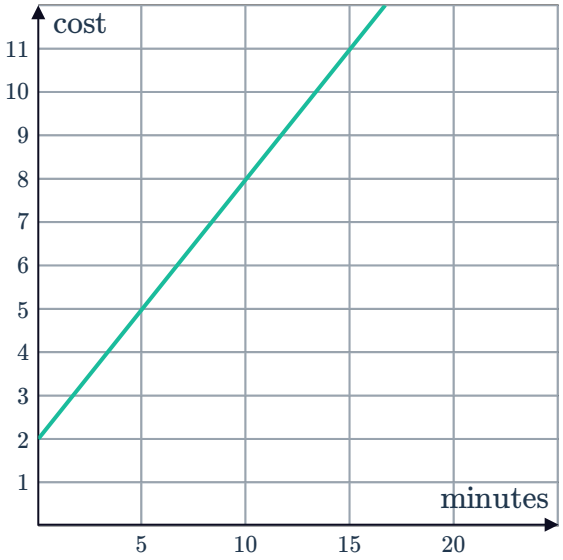
40 A bucket containing water has a hole through which the water leaks. The graph shows the amount of water remaining in the bucket after a certain number of minutes:



a Find the gradient of the line.

b Describe the meaning of the gradient in context.

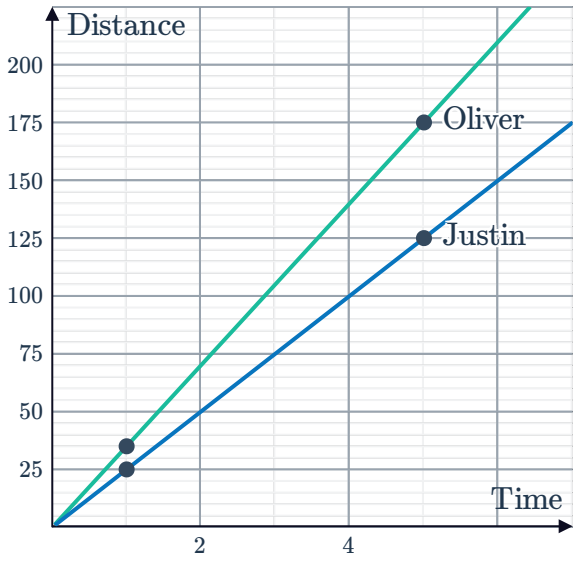
41 The graph shows the cost, in dollars, of a phone call for different call durations in minutes:



a Find the gradien of the line.

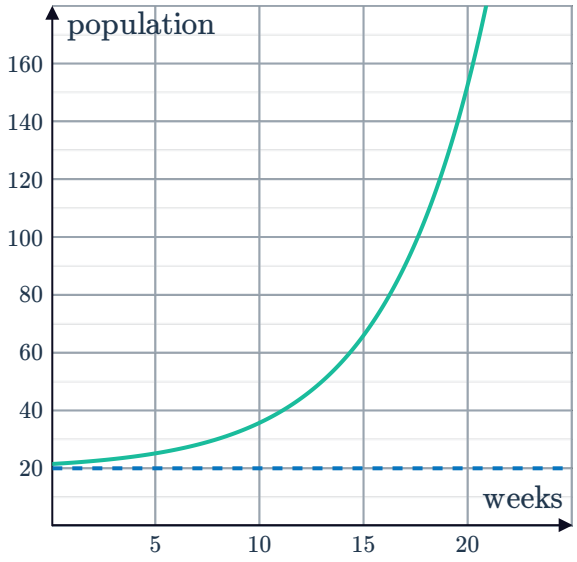
b Describe the meaning of the gradient in context.

42 The graph shows the progress of two competitors in a cycling race.
Who is travelling faster and how much faster is he travelling?

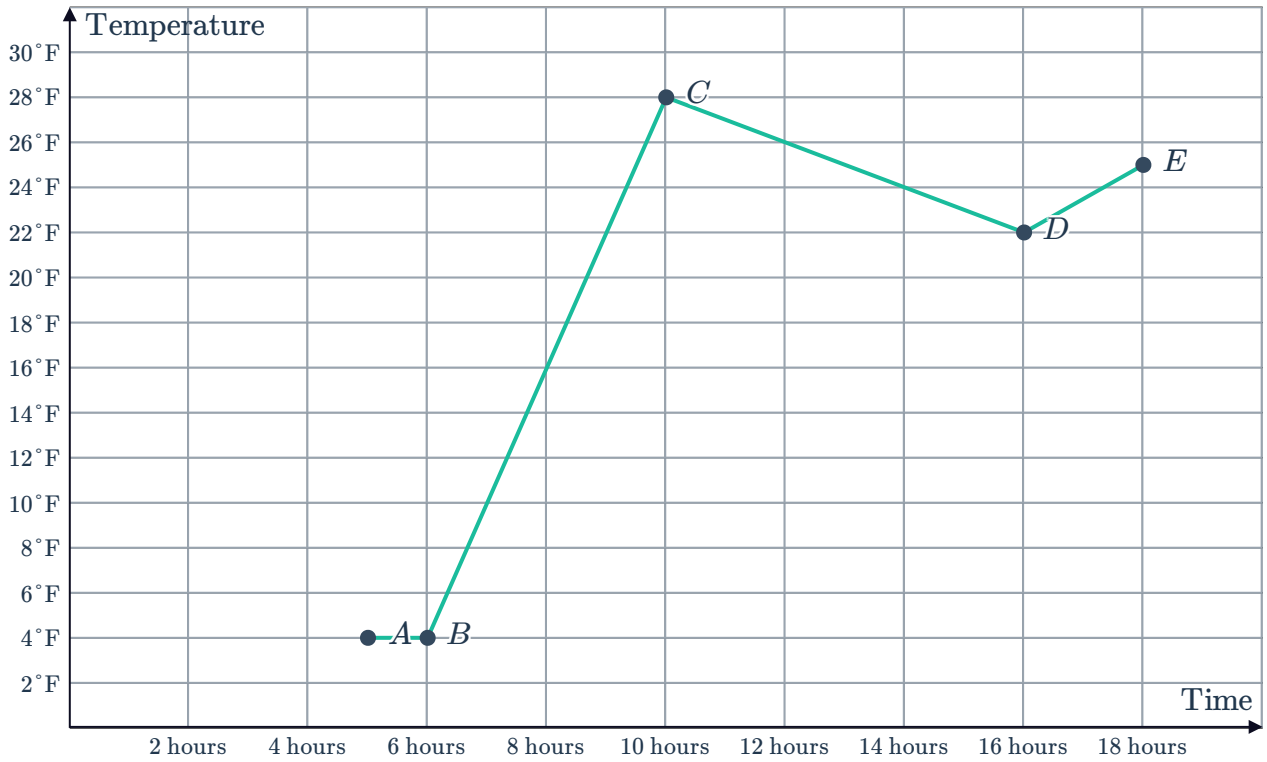


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43 The graph shows an insect population over a number of weeks:
Find the population's average rate of increase over the interval $[0, 20]$.



44 The graph shows the recorded temperatures at various hours after midnight in Antarctica:

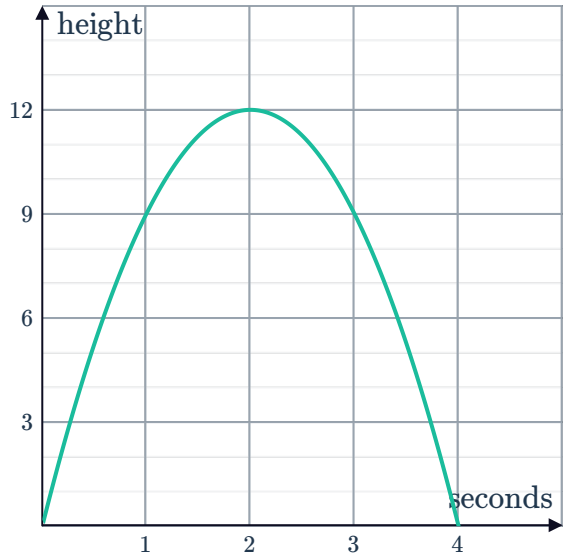


Find the rate of change in the temperature between points:

a A and B **b** B and C **c** C and D **d** D and E

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45 The graph shows the height of a cricket ball in metres after it is thrown.



a Find the rate of change of the height of the ball in the interval between:

i When it is thrown and $t = 1$.

ii $t = 1$ and when it is at its highest point.

iii When it is at its highest point and $t = 3$.

iv $t = 3$ and when it returns to the ground.

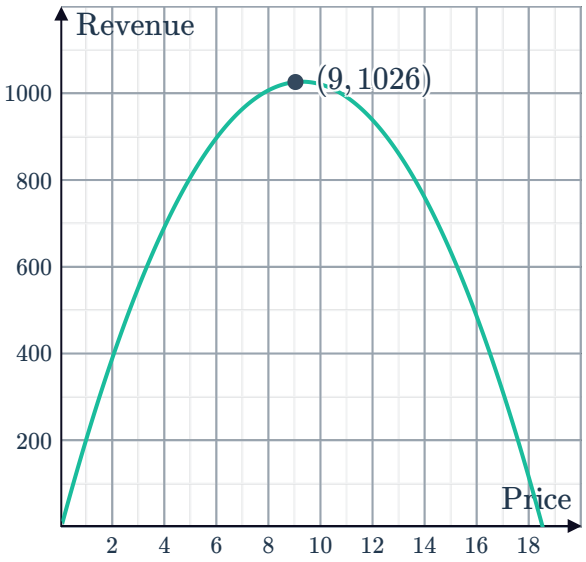
b In which of the above intervals is the ball travelling at its fastest speed?

f What do the negative rates of change in the interval between $t = 2$ and $t = 3$ and in the interval between $t = 3$ and $t = 4$ indicate?

46 The models for the revenue generated by Product A and Product B as a function of the prices they are sold at or shown:

Product A: $y = -10x^2 + 220x$

Product B:



a Find the average rate of change in the revenue generated from product A over the price interval $[0, 9]$.

b Find the average rate of change in the revenue generated from product B over the price interval $[0, 9]$.

c Which product's revenue has the hirgher rate of change over the interval $[0, 9]$?

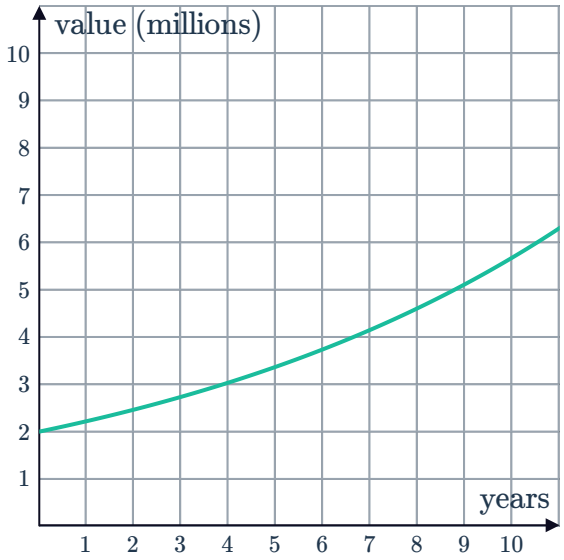
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47 Consider the value of the following two superannuation funds over ten years:

Fund A:

Year	Value (millions)
0	5.00
2	6.38
4	8.15
6	10.41
8	13.29
10	16.97

Fund B:



a Find the average rate of change in the value of Fund A over its first 10 years. Round your answer to two decimal places.

b Find the average rate of change in the value of Fund B over its first 10 years. Round your answer to two decimal places.

c Which fund performed better over its first 10 years?

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